higher education & training

Department:
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REPUBLIC OF SOUTH AFRICA

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MECHANOTECHNICS N5

(8190225)

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This question paper consists of 5 pages and a formula sheet of 3 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MECHANOTECHNICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

An epicyclic gear train has a sun wheel with 22 teeth and two planet wheels of 44 teeth each, the latter meshing with the internal teeth of a fixed annulus. The input shaft, carrying the sun wheel, transmits 4 kW at 290 r/min. The output shaft is connected to an arm that carries the planet wheels.

Calculate the following:

1.1 The number of internal teeth of the annulus (2)

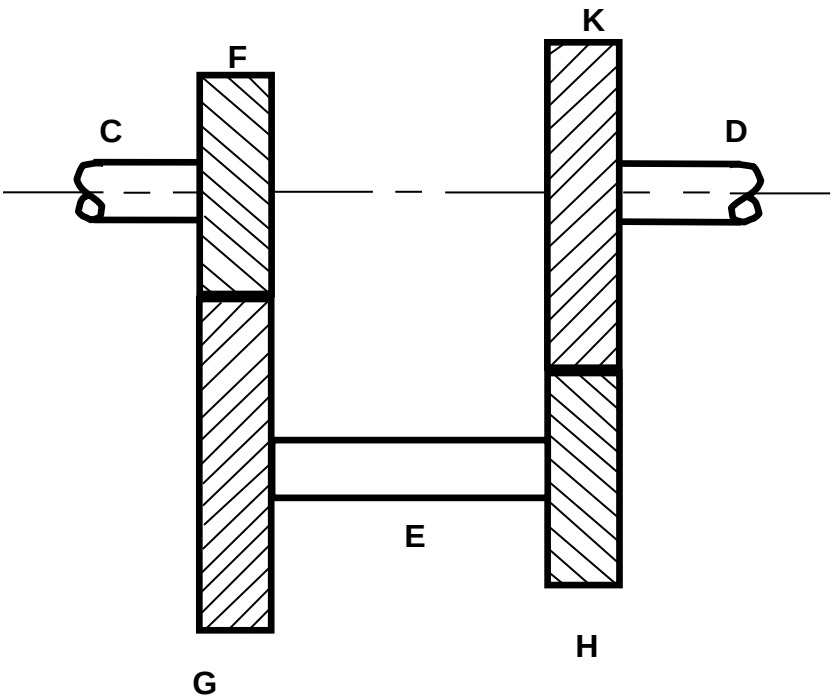
1.2 The speed of the output shaft and the torque transmitted, if the overall efficiency is 86% (10)

1.3 The speed of the annulus if it is rotated independently and the output shaft rotates at 18 r/min (4)

[16]

QUESTION 2

2.1 The FIGURE below shows two shafts, C and D, in the same line geared together through an intermediate parallel shaft E. The gears connecting shafts C and E have a module of 4 and those connecting shafts E and D have a module of 6. The pinion that meshes with gear K has 18 teeth. The speed of shaft D is one tenth of the speed of shaft C and the ratio at each reduction is the same.



FIGURE

Calculate the following:

2.1.1 The number of teeth on gears F and G (9)

2.1.2 The actual reduction (3)

2.2 State FOUR functions of a reduction gearbox. (4)

[16]

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QUESTION 3

3.1 State FOUR advantages of employing a suspended electromagnet device to remove scrap iron from a conveyor belt transporting a non-metal load. (4)

3.2 The coefficient of friction between the driving pulley and the flat belt is 0,25 and the belt width is 120 mm. The diameter of the driving pulley is 700 mm and its speed is 750 r/min. The thickness of the belt is 4 mm and the maximum allowable tension exerted on the belt must not exceed 2,5 N/mm width. The density of the belt material is $1,2 \times 10^3$ kg/m^3 and the angle of wrap between the belt and driving pulley is 160° . The efficiency of the drive is 90%.

Calculate the following:

3.2.1 The speed of the belt (2)

3.2.2 The mass of the belt (2)

3.2.3 The centrifugal force (2)

3.2.4 Tension on both sides taking centrifugal force into account (4)

3.2.5 The output power transmitted by the drive (2)

[16]

QUESTION 4

4.1 State THREE common problems that are experienced with bucket elevators and bucket conveyors. (3)

4.2 The following details apply to a bucket elevator used to transport coal:

The centre distance between the head and tail pulley	= 95 m
Bucket spacing	= 720 mm
Chain speed	= 90 m/min
Mass transported in an effective shift time of 7 hours	= 1 500 ton
The efficiency of the drive	= 82%
The discharge of the coal after top dead centre takes place at 45º	

Calculate the following:

4.2.1 The diameter of the head pulley (4)

4.2.2 The total length of the chain (assume that the top and the bottom pulleys are of the same diameter) (2)

4.2.3 The number of buckets per seconds (2)

4.2.4 The mass conveyed by each bucket (3)

4.2.5 The driving motor power required (4)

[18]

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QUESTION 5

Endless rope haulage is driven by a 70 kW motor under full-load conditions. The rope speed is 4,5 km/h and the distance between loading and discharge is 950 m. The average gradient is 1 in 10. The empty mass of the trucks is 270 kg and a mass of 680 kg is loaded in a truck. The rope has a mass of 2,5 kg/m. A force of 270 N/ton is required to overcome friction. Assume the mechanical efficiency as 75%.

Calculate the following:

5.1 Number of trucks/side (11)

5.2 Quantity of ore conveyed in an effective shift time of 7 hours (7)

[18]

QUESTION 6

An electric locomotive with a mass of 140 tons hauls trucks with a mass of 600 tons up an incline of 1 in 140. The locomotive has 80% of its mass on the driving wheels and it has a rolling resistance of 150 N/ton. The rolling resistance of the trucks is 110 N/ton.

Calculate the following:

6.1 The tractive effort of the locomotive if the coefficient of adhesion friction between the wheels and the rail is 0,3 (2)

6.2 The force required to accelerate the locomotive and the trucks up the incline (5)

6.3 The time taken for the locomotive and the trucks to accelerate from rest to 60 km/h (assume the tractive effort of the locomotive is 72% of its maximum during the accelerating period) (7)

6.4 The distance travelled during this time (2)

[16]

TOTAL: 100

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MECHANOTECHNICS N5

FORMULA SHEET

1. $m = \frac{PCD}{T}$

2. $DO = m \times (T + 2)$

3. $C = \frac{m}{2} \times (TA + TB)$

4. $Ke = \frac{1}{2} mv^2$

5. $VR = \frac{TA}{TB}$

6. $VR = \frac{PCD\ of\ gear}{PCD\ of\ pinion}$

7. $VR = \frac{NB}{NA}$

8. $NA \times TA = NB \times TB$

9. $Ft = \frac{2 \times T}{PCD}$

10. $Fr = Ft \times Tan\phi$

11. $Fn = Ft \times Sec\phi$

12. $Ie = IA + (VR)^2\ IB + (VR)^2\ IC + (VR)^2\ ID$

13. $T\alpha = Ie \times \alpha A$

14. $T\alpha = TA + \frac{(NB)}{(NA)} \frac{TBC}{\eta 1} + \frac{(ND)}{(NA)} \frac{TD}{\eta 1 \eta 2}$

15. $\frac{NB}{NA} = \frac{\omega B}{\omega A} = \frac{\alpha B}{\alpha A} = \frac{IA}{IB}$

16. $T_{OUTPUT} = T_{INPUT} \times GR \times \eta$

17. $P = \frac{\pi \times PCD}{n}$

18. $Ti + To + Th = 0$

19. $TA = TS + 2TP$

20. $\frac{Input\ speed}{Output\ speed} = \frac{Teeth\ on\ driven\ gears}{Teeth\ on\ driving\ gears}$

21. $v = \pi \times (d + t) \times N$

22. $P = Te \times v$

23. $\frac{T1}{T2} = e^{\mu\theta}$

24. $T1 = \delta \times A$

25. $Tc = m \times v^2$

26. $\frac{T1 - TC}{T2 - TC} = e^{\mu\theta cosec\ \alpha}$

27. $L = \frac{\pi}{2} \times (D + d) + \frac{(D \pm d)^2}{4 \times C} + 2C$

28. $Tg = m \times g \times sin\phi$

29. $v = \omega \times r$

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30. $v = \sqrt{\mu \times g \times r}$

31. $v = \sqrt{\frac{g \times b \times r}{2 \times h}}$

32. $v = \sqrt{gr\left[\frac{\mu + Tan\,\theta}{1 - \mu Tan\,\theta}\right]}$

33. $v = \sqrt{gr\left[\frac{hTan\theta + b/2}{h - b/2\,tan\,\theta}\right]}$

34. $\frac{T1}{T2} = \left[\frac{1 + \mu Tan\,\theta}{1 - \mu Tan\,\theta}\right]^n$

35. $Cos\,\frac{\theta}{2} = \frac{R - r}{C}$

36. $Cos\,\frac{\phi}{2} = \frac{R + r}{C}$

37. $m = w \times t \times L \times \rho$

38. $T1 = w \times n \times ft$

39. $P = Pg + P\mu$

40. $t = \frac{I \times \omega}{T}$

41. $P = \frac{2 \times \pi \times N \times T}{60}$

42. $T = F \times r$

43. $w = do + 3d - 1{,}5155P$

44. $do = de + 0{,}65P$

45. $w = \frac{\pi \times m}{2}(cos^2\,\theta)$

46. $h = m\left[1 - \frac{\pi}{4}\left(sin\,\theta\,Cos\theta\right)\right]$

47. $\frac{p1}{Rho} + \frac{(v1)^2}{2} + gh1 = \frac{p2}{Rho} + \frac{(v2)^2}{2} + gh2$

48. $Vw\,(Va) = \sqrt{\frac{gx^2}{2y}}$

49. $v = C\sqrt{mi}$

50. $hf = \frac{4 \times f \times \ell \times v^2}{2 \times g \times d}$

51. $hf = \frac{f \times \ell \times O^2}{3{,}026 \times d^5}$

52. $Q = \frac{Cd \times A \times a \times \sqrt{(2gh)}}{\sqrt{(A^2 - a^2)}}$

53. $Q = Cd \times A \times \frac{\sqrt{(2gh)}}{\sqrt{(m^2 - 1)}}$

54. $V = \sqrt{(g \times R \times Cos\theta)}$

55. Vol. bucket = $\frac{m \times s}{\rho \times v}$

56. $L = 2C + \pi D$

57. Self-weight = $\frac{m1 \times g \times S^2}{8 \times h}$

58. One load = $\frac{m2 \times g \times S}{4 \times h}$

59. T (acc load) = (T1 - T2)R

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60. T (acc drum) = $I \times \alpha = mk^2 \times \frac{a}{R}$

61. P = ω × T

62. ω = 2π × N

63. $Ke = \frac{1}{2} I \times \omega^2$

64. $Ke = \frac{work\ done}{efficiency}$

65. P = Ke × operations/sec

66. $(I_1 + I_2)\ \omega_3 = I_1 \omega_1 + I_2 \omega_2$

67. $\mu = Tan\ \theta$

68. $\eta = \frac{Tan\,\theta}{Tan\,(\theta + \phi)}$

69. T = μ × F × Re × n

70. $T = \frac{\mu \times F \times Re}{\sin\theta}$

71. T = μ × n × (Fc - S)R

72. Fc = m × ω^2 × γ

73. $Fc = \frac{mv^2}{\gamma}$

74. Tractive effort = mass on driving wheels × μ × g

75. Side thrust = $FcCos\theta - mg\ Sin\theta$

76. $\mu = \frac{FcCos\theta - mgSin\theta}{mgCos\theta + FcSin\theta}$

77. $P_l = CmgL + mgh$

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